



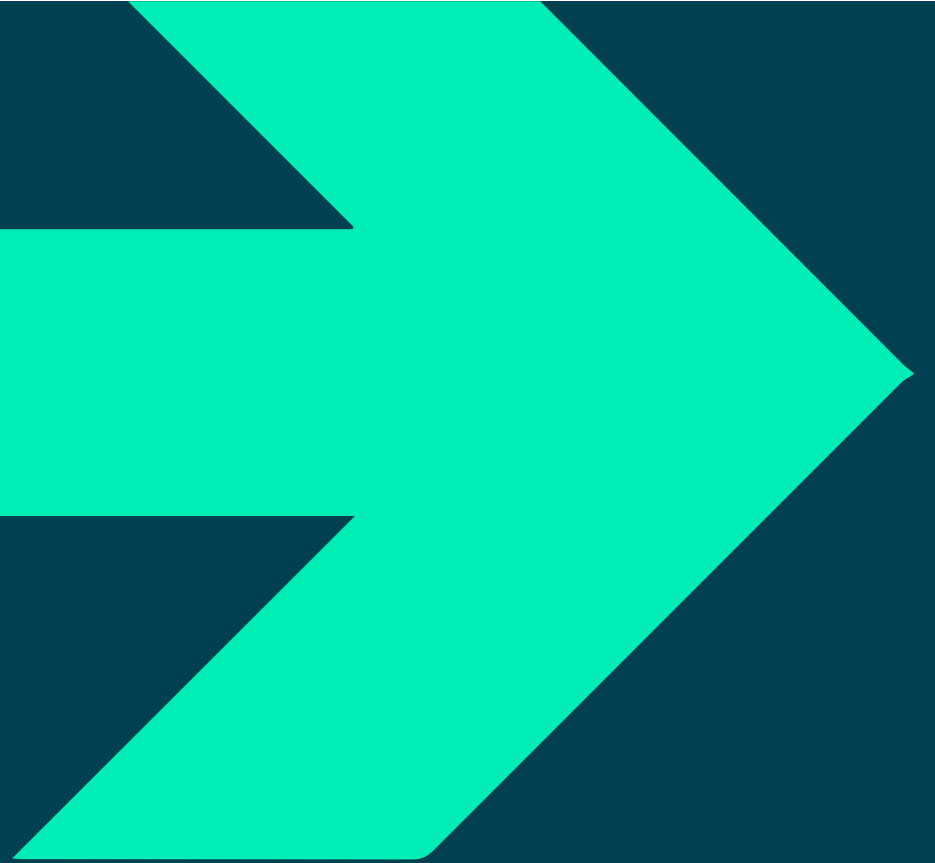
# Databases and SQL

**Tutor: Ninad Rampure**  
**Email: [Ninad.Rampure@qa.com](mailto:Ninad.Rampure@qa.com)**





# Data Normalisation





# THE NEED FOR NORMALISATION

## **Some Data has a Simple Structure:**

- Customer
- Name, Address, Telephone, Balance

## **Some Data has a Complex Structure:**

- But what if Customer has more than one address?
- Home phone, Mobile, business phone....





# CUSTOMERS AND ORDERS

**QA Games**

22 Gamesville Road,  
Gamesville,  
GM1 8GG

**To:**

Games-R-Us,  
45 Shoppington Ave.  
Shoppington  
Bucks,  
SH8 2AB

**ORDER FORM**

**Order No:**100001

**Date:** 31/07/2011

**Customer No:** 8573

| Quantity | Product ID | Description                     | Unit Price       | Amount         |
|----------|------------|---------------------------------|------------------|----------------|
| 2        | 849        | Alpine Ski Instructor           | 29.99            | £59.98         |
| 28       | 773        | SIM Accountant                  | 23.46            | £656.88        |
| 6        | 382        | Recorder Hero – Legends of Folk | 19.99            | £119.94        |
|          |            |                                 |                  |                |
|          |            |                                 |                  |                |
|          |            |                                 |                  |                |
|          |            |                                 | <b>Total Due</b> | <b>£836.80</b> |



# CUSTOMERS AND ORDERS EXCEL STYLE

How many copies of Alpine Ski Instructor  
did *Games-R-Us* order in 2011

|   | A        | B          | C         | D           | E             | F          | G            | H              |          | J                | K          | L           | M            | N          |
|---|----------|------------|-----------|-------------|---------------|------------|--------------|----------------|----------|------------------|------------|-------------|--------------|------------|
| 1 | Ord. No. | Date       | Cust. No. | Cust. Name. | Cust. Address | Cust. Town | Cust. County | Cust. PostCode | Prod. ID | Prod. Desc.      | Prod. Qty. | Prod. Price | Prod. Amount | Ord. Total |
| 2 | 100001   | 31/07/2011 | 8573      | Games-R-Us  | 45 Shoppingt  | Shoppingtc | Bucks.       | SH8 2AB        | 849      | Alpine Ski Instr | 2          | £29.99      | £59.98       | £836.80    |
| 3 |          |            |           |             |               |            |              |                | 773      | SIM Accountan    | 28         | £23.46      | £656.88      | £836.80    |
| 4 |          |            |           |             |               |            |              |                | 382      | Recorder Hero    | 6          | £19.99      | £119.94      | £836.80    |
| 5 | 100002   | 31/07/2011 | 5644      | Gamerz      | 304 The Mall  | Mallville  | Herts.       | MV9 1AQ        | 849      | Alpine Ski Instr | 200        | £29.99      | £5,998.00    | £6,772.50  |
| 6 |          |            |           |             |               |            |              |                | 562      | Deck Chair Atti  | 50         | £15.49      | £774.50      | £6,772.50  |
| 7 |          |            |           |             |               |            |              |                |          |                  |            |             |              |            |

What are total sales of Deck Chair  
Attendant Deluxe in the county of  
Buckinghamshire?



# NORMALISATION

## **Normalisation method:**

- Formal and mechanical approach
- Rigorous, unlike Entity-Relationship method
- Useful for transforming existing files
- Check existing databases, validate extensions
- Lossless Decomposition

## **Benefits:**

- Strategy for key and table selection
- Improved ad-hoc queries and interfaces
- Eliminates insertion anomalies
- Eliminates deletion anomalies
- Reduction in DB modification time
- Correct representation of physical database
- Problems show up early in the design



## 3 TYPES OF DATA ANOMALY

- **Update Anomalies** happen when the person charged with the task of keeping all the records current and accurate, is asked, for example, to change an employee's title due to a promotion. If the data is stored redundantly (repeated) in the same table, and the person misses any of them, then there will be multiple titles associated with the employee. The end user has no way of knowing which is the correct title.
- **Insertion Anomalies** happen when inserting vital data into the database is not possible because other data is not already there. For example, if a system is designed to require that a customer be on file before a sale can be made to that customer, but you cannot add a customer until they have bought something, then you have an insert anomaly. It is the classic "catch-22" situation.
- **Deletion Anomalies** happen when the deletion of unwanted information causes desired information to be deleted as well. For example, if a single database record contains information about a particular product along with information about a salesperson for the company and the salesperson quits, then information about the product is deleted along with salesperson information.



## STEPS TO ACHIEVE 3NF SUMMARY

### 1NF - No repeating columns

- Add extra columns
- Data should have atomicity

### 2NF + all nonkey attributes must be dependent on the whole key

- Identify the key
- Identify **Functional Dependencies** (FDs)
- Identify columns that depend on only part of the key
- Create new tables for these columns and partial key

### 3NF + all nonkey attributes must only be dependent on the key

- Identify **Transitive Dependencies** on nonkey attributes
- Create new tables for those columns





# TRANSITIVE DEPENDENCY

5

## Example 1: Determine NF

- ◆ ISBN → Title
- ◆ ISBN → Publisher
- ◆ Publisher → Address

Publisher is a non-key attribute, and it determines Address, another non-key attribute. Therefore, there is a transitive dependency, which means that the relation is NOT in 3 NF.

BOOK

|             |       |           |         |
|-------------|-------|-----------|---------|
| <u>ISBN</u> | Title | Publisher | Address |
|-------------|-------|-----------|---------|

8

ISBN uniquely identifies a specific title and edition of a book, with different ISBNs assigned to distinct formats (like hardcover or paperback), different languages, and different versions of the same title.